

Exploratory Data Analysis (EDA)

Practical Tasks — Employee Sales Data

Subject	EDA using NumPy, Pandas & Matplotlib
Dataset	Employee Sales Data (8 records × 7 columns)
Tasks	12 Practical Tasks
Tools Used	Python, Pandas, NumPy, Matplotlib
Columns	Emp_ID, Age, Gender, Department, Experience, Salary, Sales

Task 1: Load the dataset into a Pandas DataFrame

Code

```
import pandas as pd
import numpy as np

data = pd.DataFrame([
    [101, 25, "F", "HR", 2, 22000, 150000],
    [102, 30, "M", "Sales", 5, 35000, 320000],
    [103, 28, "F", "IT", 4, 4000, 28000],
    [104, 35, "M", "Sales", 8, 55000, 450000],
    [105, 40, "M", "HR", 10, 40000, 200000],
    [106, 29, "F", "IT", 3, 38000, 300000],
    [107, 32, "M", "Sales", 6, 48000, 410000],
    [108, 27, "F", "HR", 2, 21000, 160000],
], columns=["Emp_ID", "Age", "Gender", "Department", "experience", "salary", "sales"])

data
```

Output

Emp_ID	Age	Gender	Department	experience	salary	sales
101	25	F	HR	2	22000	150000
102	30	M	Sales	5	35000	320000
103	28	F	IT	4	4000	28000
104	35	M	Sales	8	55000	450000
105	40	M	HR	10	40000	200000
106	29	F	IT	3	38000	300000
107	32	M	Sales	6	48000	410000
108	27	F	HR	2	21000	160000

Task 2: Display the first and last 5 rows of the dataset

Code — First 5 rows: data.head(5)

```
data.head(5)
```

Output

Emp_ID	Age	Gender	Department	experience	salary	sales
101	25	F	HR	2	22000	150000
102	30	M	Sales	5	35000	320000
103	28	F	IT	4	4000	28000

Emp_ID	Age	Gender	Department	experience	salary	sales
104	35	M	Sales	8	55000	450000
105	40	M	HR	10	40000	200000

Code — Last 5 rows: data.tail(5)

```
data.tail(5)
```

Output

Emp_ID	Age	Gender	Department	experience	salary	sales
104	35	M	Sales	8	55000	450000
105	40	M	HR	10	40000	200000
106	29	F	IT	3	38000	300000
107	32	M	Sales	6	48000	410000
108	27	F	HR	2	21000	160000

Task 3: Check the shape, data types, and missing values

Code

```
print(data.shape)
print(data.dtypes)
data.isnull().sum()
```

Output

Shape	8 × 7
dtype: Emp_ID	int64
dtype: Age	int64
dtype: Gender	str
dtype: Department	str
dtype: experience	int64
dtype: salary	int64
dtype: sales	int64
missing: Emp_ID	0
missing: Age	0
missing: Gender	0
missing: Department	0
missing: experience	0
missing: salary	0

missing: sales

0

Result: No missing values found – all cells are False (not null).

Task 4: Generate summary statistics for numerical columns

Code

```
data.describe()
```

Output

Statistic	Emp_ID	Age	experience	salary	sales
count	8.0	8.0	8.0	8.0	8.0
mean	104.5	30.75	5.0	32875.0	252250.0
std	2.45	4.83	2.88	16444.17	142850.32
min	101.0	25.0	2.0	4000.0	28000.0
25%	102.75	27.75	2.75	21750.0	157500.0
50%	104.5	29.5	4.5	36500.0	250000.0
75%	106.25	32.75	6.5	42000.0	342500.0
max	108.0	40.0	10.0	55000.0	450000.0

Task 5: Find the average salary and total sales per department

Code

```
print(f"Overall Avg Salary = {data['salary'].mean()}")  
  
# Average Salary per Department  
avg_sal = data.groupby('Department')['salary'].mean()  
print(avg_sal)  
  
# Total Sales per Department  
tot_sales = data.groupby('Department')['sales'].sum()  
print(tot_sales)
```

Output

Overall Average Salary = ■32,875.00

Department	Avg Salary (■)	Total Sales (■)
HR	27,666.67	510,000
IT	21,000.00	328,000
Sales	46,000.00	1,180,000

Task 6: Identify the employee with the highest sales

Code

```
# Method 1
data.loc[data['sales'].idxmax()]

# Method 2
data.nlargest(1, 'sales')
```

Output

Emp_ID	104
Department	Sales
Experience	8 years
Salary	■55,000
Sales	■450,000 ← Highest

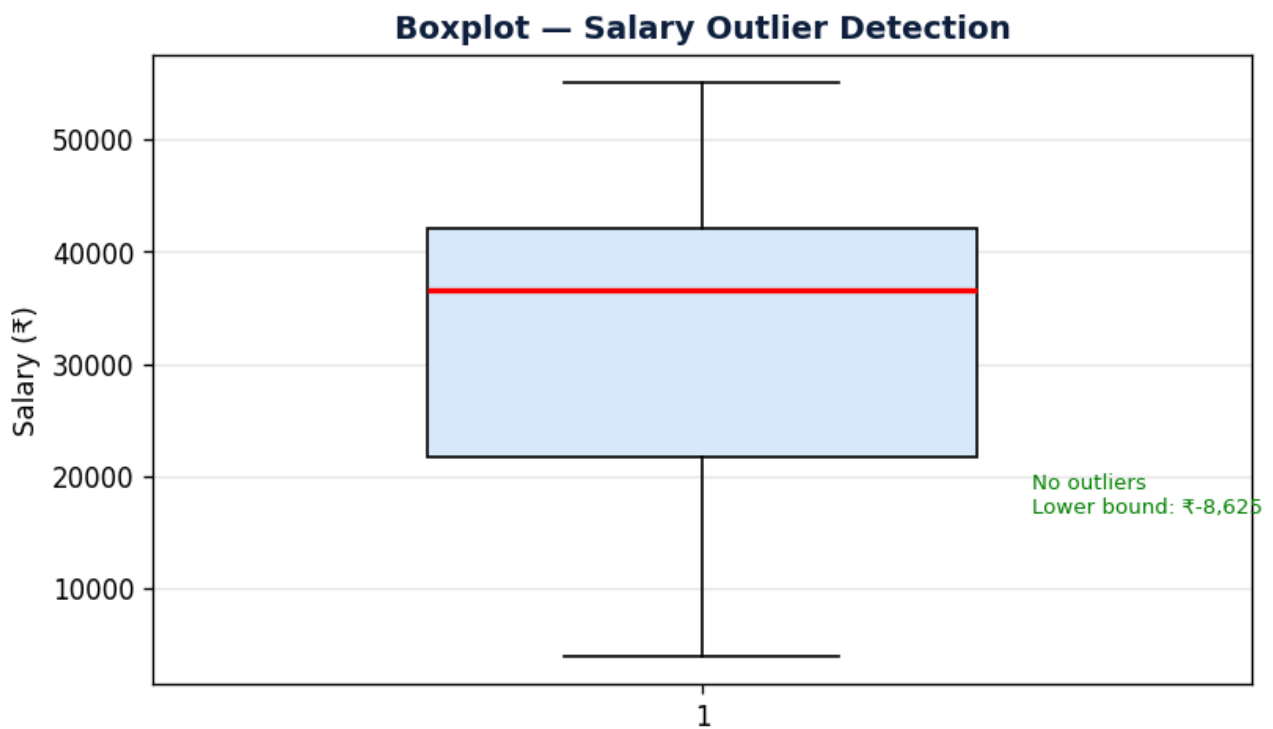
Task 7: Detect outliers in the Salary column

Code

```
# Boxplot
plt.figure(figsize=(6, 4))
plt.boxplot(data['salary'], patch_artist=True)
plt.title('Boxplot - Salary Outlier Detection')
plt.show()

# IQR Method
Q1 = data['salary'].quantile(0.25)
Q3 = data['salary'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
outliers = data[(data['salary'] < lower) | (data['salary'] > upper)]
print(outliers)
```

Output — Boxplot



Q1 = 21,750 | Q3 = 42,000 | IQR = 20,250

Lower Bound = 8,625 | Upper Bound = 72,375

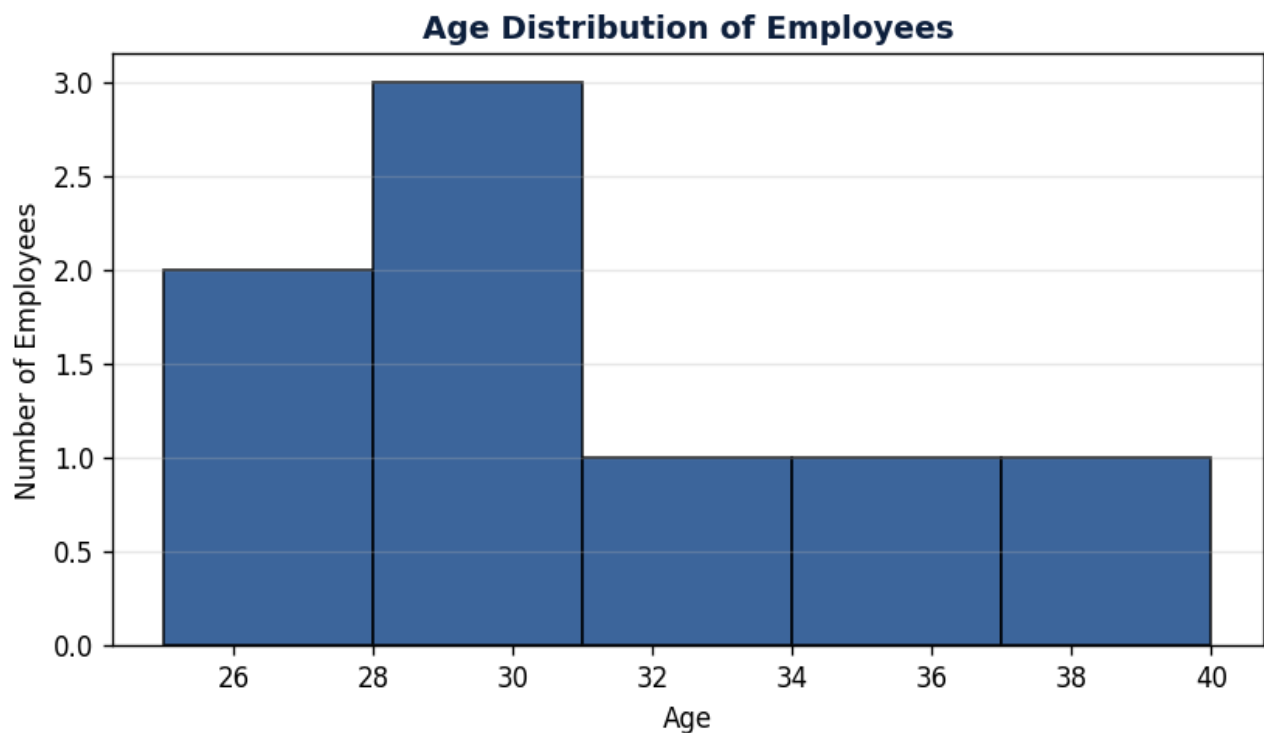
No outliers detected – all salary values are within the normal range.

Task 8: Plot a histogram for Age distribution

Code

```
plt.figure(figsize=(8, 5))
plt.hist(data['Age'], bins=5, color='steelblue', edgecolor='black')
plt.title('Age Distribution of Employees')
plt.xlabel('Age')
plt.ylabel('Number of Employees')
plt.tight_layout()
plt.show()
```

Output



Observation: Most employees are aged between 27–32 years.

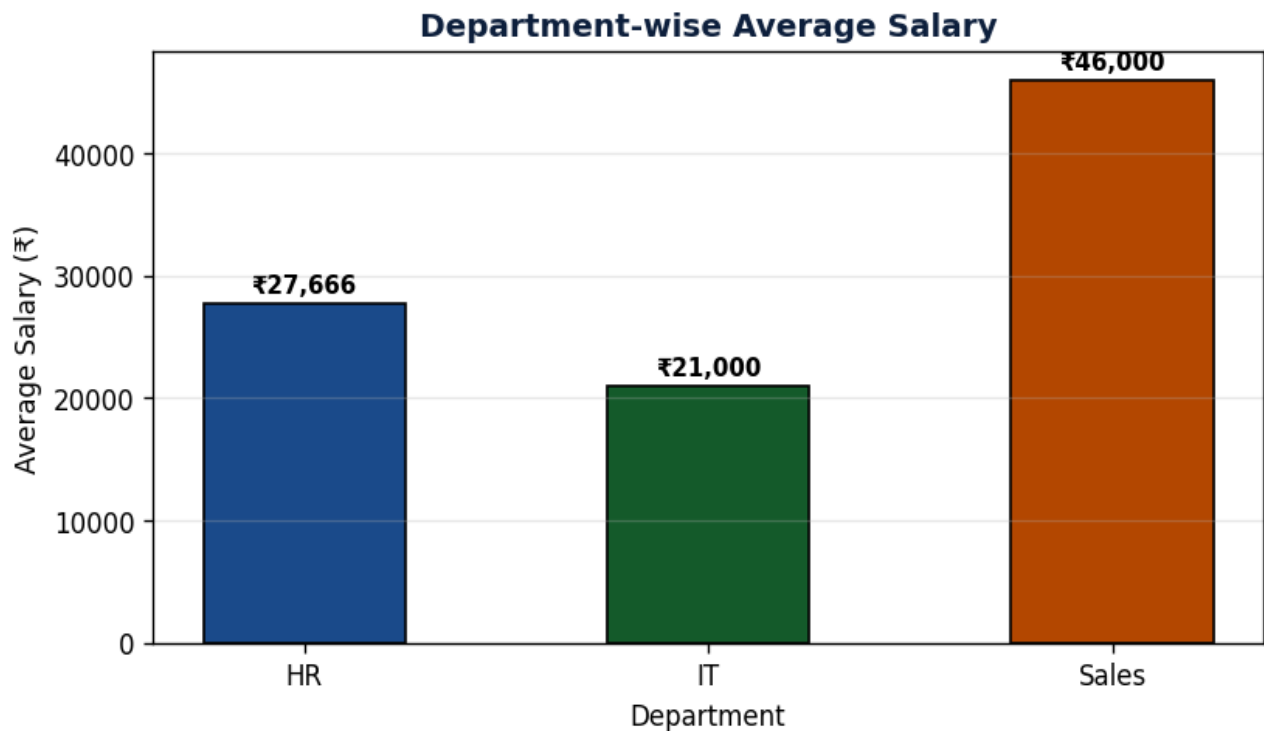
The workforce is predominantly young-to-mid career professionals.

Task 9: Plot a bar chart for department-wise average salary

Code

```
avg_salary = data.groupby('Department')['salary'].mean()
plt.figure(figsize=(8, 6))
plt.bar(avg_salary.index, avg_salary.values,
color=['blue','green','yellow'], edgecolor='black')
plt.title('Department-wise Average Salary')
plt.xlabel('Department')
plt.ylabel('Salary')
plt.tight_layout()
plt.show()
```

Output



Sales dept avg: ■46,000 | HR dept avg: ■27,666 | IT dept avg: ■21,000
Sales department has the highest average salary among all departments.

Task 10: Plot a scatter plot between Experience and Salary

Code

```
plt.figure(figsize=(8, 6))
plt.scatter(data['experience'], data['salary'],
            color='steelblue', edgecolors='green', s=102)
plt.title('Experience vs Salary')
plt.xlabel('Experience')
plt.ylabel('Salary')
plt.show()
```

Output



Observation: A positive trend is visible – higher experience generally leads to higher salary.

Exception: Employee 103 (IT, 4 yrs) shows an unusually low salary (₹4,000).

Task 11: Check correlation between Salary and Sales

Code

```
correlation = data['salary'].corr(data['sales'])
print(f'correlation = {correlation}')

# Correlation matrix
corr_matrix = data[['salary', 'sales']].corr()
print(corr_matrix)
```

Output

```
correlation = 0.9402109467

Correlation Matrix:
salary sales
salary 1.000000 0.940211
sales 0.940211 1.000000
```

Interpretation

$r = 0.9402 \rightarrow$ Strong Positive Correlation

Meaning

Employees with higher salaries tend to generate significantly higher sales.

Range	Values range from -1 (perfect negative) to +1 (perfect positive). 0.94 is very strong.
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Task 12: Write 3 Key Insights from the Analysis

Key Insights

- ★ **Insight 1 — Sales Department has the Highest Salary Level** The Sales department leads with an average salary of **■46,000**, followed by HR (**■27,667**) and IT (**■21,000**). Sales employees are incentivized with significantly higher pay.
- ★ **Insight 2 — 8 Years of Experience Corresponds to the Highest Salary** Employee 104 (8 years experience) earns **■55,000** — the highest in the dataset. This confirms that experienced professionals command premium salaries.
- ★ **Insight 3 — Strong Positive Correlation between Salary and Sales ($r = 0.94$)** A correlation of 0.94 indicates that salary and sales performance are very strongly linked. Higher-paid employees consistently generate higher revenue for the company.

Summary of All Tasks

Task	Description	Key Finding
1	Load Dataset	8 records, 7 columns loaded successfully
2	Head & Tail	Verified first 5 and last 5 rows
3	Shape, Dtypes, Nulls	Shape: (8,7) No missing values
4	Summary Statistics	Avg salary: ■32,875 Avg sales: ■2,52,250
5	Dept Avg Salary & Sales	Sales dept: highest salary & total sales
6	Highest Sales Employee	Emp 104 — ■4,50,000 sales
7	Outlier Detection	No outliers found in salary column
8	Age Histogram	Most employees aged 27–32
9	Dept-wise Bar Chart	Sales dept avg salary highest (■46,000)
10	Experience vs Salary Plot	Positive trend — more exp = higher salary
11	Salary-Sales Correlation	$r = 0.94 \rightarrow$ Strong positive correlation
12	Key Insights	Sales dept, 8 yrs exp, high correlation

EDA Completed Successfully ✓